Paleomagnetism and geochronology of sills of the Doherty Mountain area, southwestern Montana: Implications for the timing of fold-and-thrust belt deformation and vertical-axis rotations along the southern margin of the Helena salient

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ABSTRACT

Paleomagnetic and $^{40}Ar/^{39}Ar$ data from intermediate composition sills exposed along the southern margin of the Helena salient in southwestern Montana help refine the age of Late Cretaceous contraction in this part of the North American Cordillera. The structures in the Doherty Mountain fold complex deform Cambrian through Mississippian strata in the hanging wall of the Jefferson Canyon–Lombard fault system. Sills exposed within these strata, which are complexly deformed into northeast- to northwest-plunging anticlines and a northwest-plunging syncline give well-defined remanent magnetizations and, after correction for fold plunge followed by bedding tilt, paleomagnetic data from 20 of 21 sites from four folds yield positive fold tests at 100 percent unfolding. The magnetizations from these sites are interpreted to be primary remanent magnetizations acquired during sill emplacement prior to fold-and-thrust deformation, although prefolding remagnetization due to widespread latest Cretaceous magmatism in the area cannot be ruled out. One site yields a magnetization that appears to postdate folding; this site has either been remagnetized or is in a distinct intrusive phase emplaced subsequent to folding. Biotite mineral separates from sills from two folds yield identical $^{40}Ar/^{39}Ar$ plateau dates of 77 Ma. Although the group mean directions from sills in each of the folds may represent incomplete samplings of the geomagnetic field and the number of independent readings is low, their grand-mean direction (declination [Dec.] = 347°, inclination [Inc.] = 66°, $k = 71.9$, $\alpha_{95} = 9°$, N = 5 independent observations based on results from 21 sites) is similar to the expected Late Cretaceous reference direction for the study area, based on results from the essentially coeval Adel Mountain Volcanics. Therefore, these data suggest that deformation began after 77 Ma (Campanian-Maestrichtian) and that upper plate strata in the Lombard thrust sheet of the Helena salient in this area have not experienced significant local vertical-axis rotation due to thrust movement and impingement with foreland structures along the Southwest Montana Transverse Zone. The absence of appreciable vertical-axis rotations is consistent with data obtained from other paleomagnetic studies in the region.

Keywords: paleomagnetism, geochronology, tectonics, structural geology, thrust belt deformation.

INTRODUCTION

The geology of southwestern Montana is characterized by two distinct styles of latest Cretaceous to early Tertiary deformation: (1) thin-skinned folds and thrust faults (sevier style) of the Helena salient of the Cordilleran thrust belt; and (2) basement-involved folds, reverse faults, and thrust faults (Laramide style) of the Rocky Mountain foreland (Fig. 1) (Schmidt et al., 1990). Locally, the areas affected by these two contrasting styles of deformation overlap. Both deformation styles were approximately coeval, based on crosscutting relationships of folds and faults, but estimates of absolute ages of deformation are rare due to a lack of synorogenic sedimentary rocks that can be used to limit the age of compressional deformation (e.g., Harlan et al., 1988; Kulik and Schmidt, 1988; Perry et al., 1988; Schwartz and DeCelles, 1988).

In the Doherty Mountain area (Fig. 2), intermediate composition sills are exposed in a set of northwest- to northeast-plunging folds. Regional and local geologic mapping indicates that folding accompanied strong internal shortening (~65%) of the Lombard thrust sheet at its southern limit (Burton et al., 1995; Bergin and Whisner, 1995). Latite to andesitic sills have pervasively intruded Cambrian through Mississippian strata and are well exposed in Cottonwood Canyon, which transects the southern part of the study area. The age of the sills has not been determined by isotopic methods, but they have been mapped as Late Cretaceous or Eocene in age (Robinson, 1963; Vuke et al., 2002). Although the sills are concordant with fold structures, the relationship between sill emplacement and folding is unknown. Harlan et al. (1988) and Harlan et al. (1995) showed that

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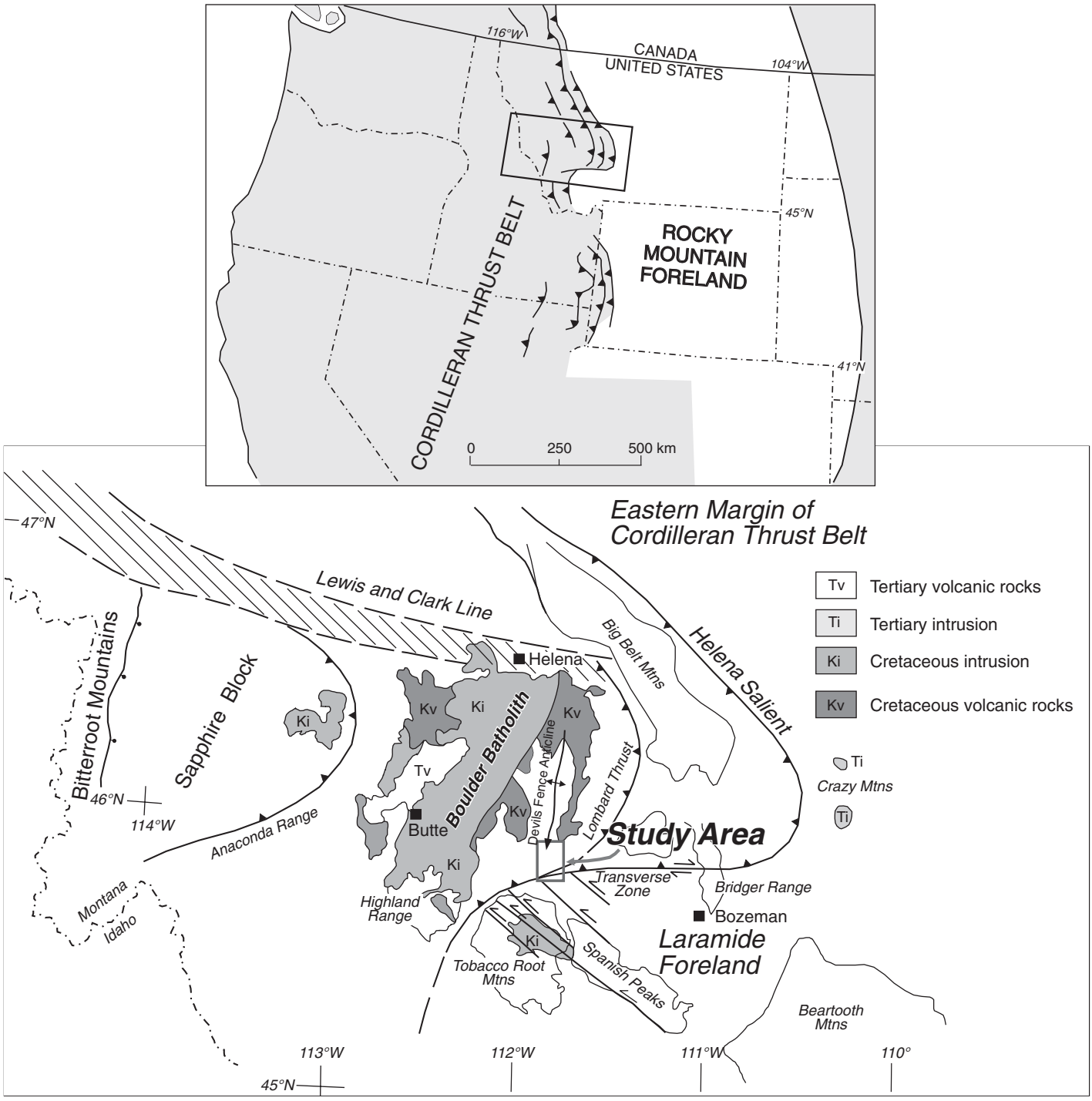


Figure 1. Map showing the tectonic setting of the Helena salient of the Montana part of the North American Cordilleran orogenic belt. Also shown is the location of the study area in the transverse zone along the southern margin of the salient. Figure modified from Burton et al. (1995).

sill emplacement in folds sometimes postdates folding and that the apparent fold geometry is an artifact of sills intruding along favorably oriented bedding planes in preexisting folds. However, given that syndeformational intrusive and volcanic rocks of latest Cretaceous age are widespread within and near the Helena salient (Schmidt et al., 1990; Lageson et al., 2001; Kalakay et al., 2001) and adjacent foreland structures, the exact relationship between sill intrusion and folding remains to be determined. Our research tests the hypothesis that the Doherty Mountain sills in Cottonwood Canyon and surrounding areas have been partly to fully folded during thin-skinned deformation in this area. Resolving the age of sill intrusion with respect to folding could thus provide important information regarding the age of fold and thrust belt deformation within the leading edge of the Cordilleran fold and thrust belt in this part of Montana.

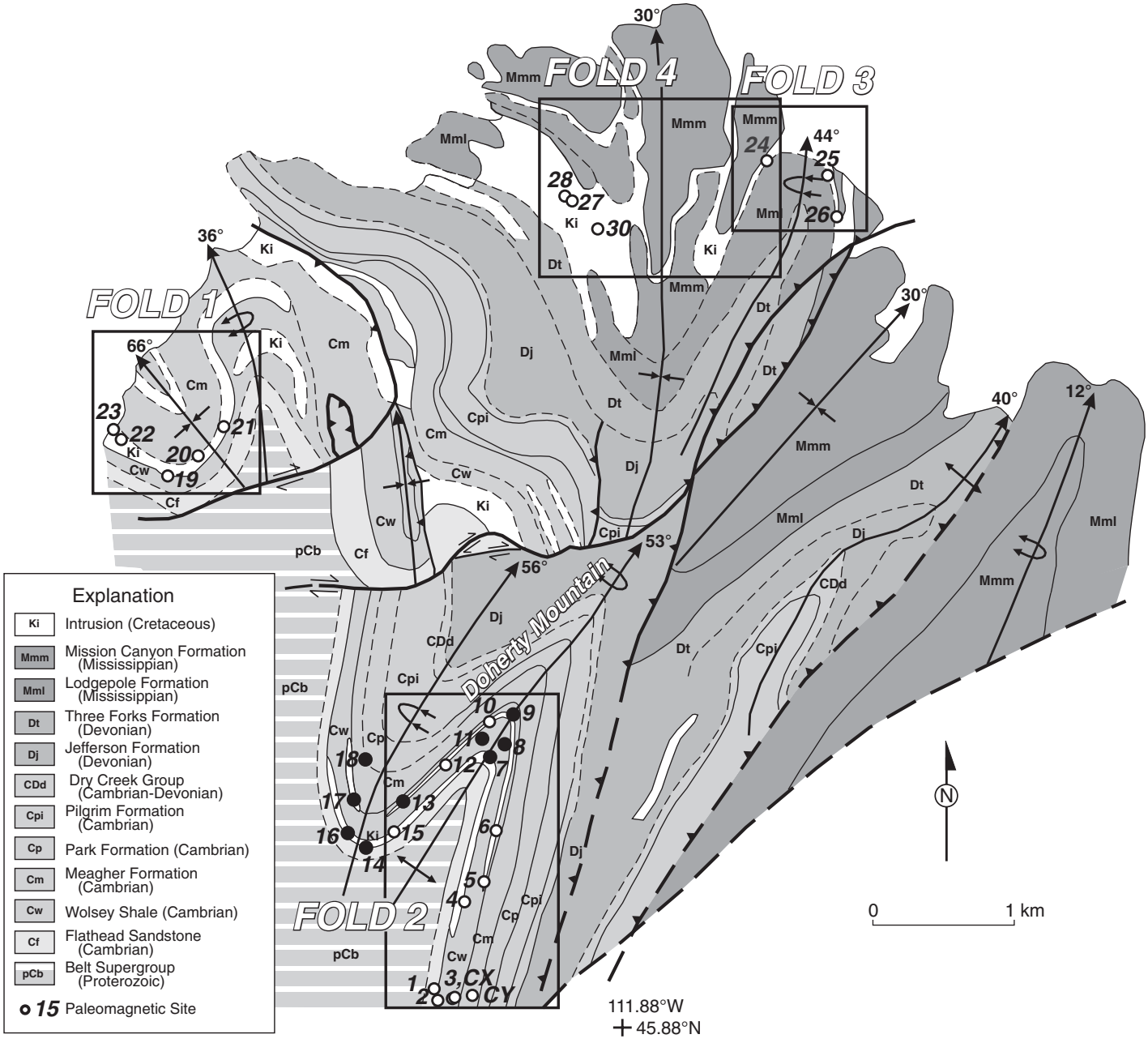


Figure 2. Geologic map of the Doherty Mountain area, southwest Montana. Also shown are paleomagnetic sites sampled during the study and the four folds for which paleomagnetic fold tests were conducted. Open circles are sites that yielded usable paleomagnetic data; closed circles are sites that failed to yield usable paleomagnetic results. Boxes and labeled folds are those for which paleomagnetic fold tests were conducted as described in the text. Figure based on unpublished geologic mapping by S.C. Whisner and C.J. Schmidt (1998).

GEOLOGIC SETTING

The Doherty Mountain area is located from 5 to 12 km north of the southern margin of the Helena salient, a large convex-eastward bend in the Cordilleran fold and thrust belt (Fig. 1). The southern boundary of the Helena salient is composed of a system of northeast- to east-striking faults collectively called the Southwest Montana Transverse Zone (Schmidt et al., 1988). The principal thrust fault of the Helena salient is the Lombard thrust and the main fault of the Southwest Montana Transverse Zone is the Jefferson Canyon–Cave fault system. The Lombard thrust is inferred to connect with the Jefferson Canyon–Cave fault system, defining the eastern and southern boundaries of the Lombard thrust sheet. Folds of the Doherty Mountain area (Fig. 2) are a manifestation of internal shortening in the Lombard sheet and are inferred to have been coeval with movement on the Lombard thrust and Jefferson Canyon–Cave fault system.

Rock units ranging in age from the Proterozoic Belt Supergroup (LaHood Formation) to the Mississippian Madison Group and the concordant sills are tightly folded in nearly similar style, with thinned limbs and thickened hinges

especially in the less competent shales and the sills. Shortening in the Doherty Mountain area is inferred to have occurred initially by detachment folding above the Lombard thrust, followed by thrust faulting across fold hinges and limbs. Folds or fold pairs were thrust over other fold pairs, eliminating intervening folds (Whisner, 1998). East-west shortening across the area is estimated at 55 percent (Whisner, 1998).

Fold axes in the Doherty Mountain area, herein termed the Doherty Mountain fold complex (Fig. 2), plunge to the north, with an average of 41 degrees. This north-plunge of the folds is inferred to reflect the shape of the underlying Lombard thrust. To the north the thrust is inferred to have a southerly dip that is manifested in the southern plunge of the Devils Fence Anticline, a major culmination in the Lombard thrust sheet (Fig. 1). In the Doherty Mountain area, the thrust is inferred to dip north into a major sag or depression in the sheet (Whisner, 1998). Azimuths of fold axes fan from northwest to northeast. This fanning of fold axes may be the result of imbrication, in which the fold axes of early formed folds were rotated back as the folds were faulted over younger folds and faults formed beneath them. Alternatively, the distribution of fold axes azimuths may reflect systematic vertical-axis rotations of upper plate rocks as the southern margin of the Lombard thrust sheet is approached, a hypothesis that is testable using appropriate paleomagnetic data.

FIELD AND LABORATORY METHODS

To determine the relationship between sill intrusion and folding in the Helena salient, we collected samples for paleomagnetic analysis at 30 sites exposed in five folds in the Doherty Mountain fold complex (Fig. 2) (Whisner, 1998). The southernmost fold (Fold 2, Fig. 2) consists of two oppositely verging asymmetric anticlines that plunge 53° and 56° northeast. For this fold pair, we collected samples from three different sills in the Cambrian Flathead and Woolsey formations for a total of 18 sites. We sampled four sites around a northwest-plunging syncline (Fold 1, Fig. 2) in a sill that intrudes the Cambrian Woolsey Formation. Along the northern part of the study area, we sampled sills that intrude Mississippian Lodgepole Limestone in two folds (Folds 3 and 4, Fig. 2) that plunge north. Samples were collected in sills on both limbs of the folds and, where possible, fold hinges, in order to conduct fold tests to determine the age of sill remanence relative to folding. In order to assess the long-term geologic stability of the natural remanent magnetization of the sills, we collected samples in host carbonate strata from the baked contact zone of one sill and from a site believed to be well removed from the potential thermal effects of sill intrusion. Samples were collected for $^{40}Ar/^{39}Ar$ dating from two locations in order to precisely determine the age of sill emplacement.

Samples were collected as cores using a portable drill equipped with a diamond-cored bit, with 8–12 individually oriented samples obtained at each site. Sample orientations were determined using magnetic and solar compasses prior to core extraction. In the laboratory, at least one specimen from each sample was subjected to progressive alternating field (AF) demagnetization, usually to above 100 mT. A subset of samples was subjected to thermal demagnetization. Characteristic remanent magnetization (ChRM) components were determined by visual inspection of orthogonal vector diagrams and stereographic projections, with principal components analysis (Kirschvink, 1980) used to identify the ChRM directions. In cases where a stable endpoint, defined by three or more points defining a well-defined linear demagnetization interval, could not be fully isolated during either AF or thermal demagnetization due to overlapping components of magnetization, great circles were fit to curvilinear demagnetization trajectories. For sites dominated by stable endpoint behavior, site-mean directions were estimated using Fisher (1953) statistics; for sites in which the ChRM could not be isolated for all samples, a combination of stable endpoint and great circle analysis (McFadden and McElhinny, 1988) was used to determine the best estimate of the site-mean directions.

The magnetic carriers of remanence were determined using a combination of paleomagnetic and rock magnetic methods, including acquisition of a saturation isothermal remanent magnetization (SIRM), AF demagnetization of natural remanent magnetization, AF demagnetization of an anhysteretic magnetization (ARM) acquired in a field of 0.1 mT, backfield demagnetization of the SIRM, and thermal demagnetization of three component IRM (Lowrie, 1990). In addition, characteristics of the laboratory unblocking temperatures of the ChRM as well as magnetic susceptibility versus temperature analysis were also used to assess the remanence carriers.

Because the folds in this study have substantial plunge and it has been recognized that failure to account for moderate to steep plunges in folds can influence results from paleomagnetic analysis (e.g., MacDonald, 1980; Weil and Van der Voo, 2002), we conducted our fold tests in the following manner. For each of the four fold tests, we first corrected the in situ paleomagnetic site-mean directions for the individual fold plunge determined by π-diagram analysis for each fold. This correction was followed by incrementally restoring the site-mean directions to the assumed paleohorizontal based on the strike and dip of the host strata. The statistical significance of the fold tests was determined using Watson and Enkin's (1993) optimal concentration method and Enkin's (2003) direction-correlation (D.C.) method.

Samples from two sills contain biotite of sufficient quality to be appropriate for $^{40}Ar/^{39}Ar$ dating. Two biotite mineral concentrates from samples collected at paleomagnetic sites PMD1 (Fold 2) and PMD20 (Fold 1) (Fig. 2) were prepared from hand samples using standard gravimetric (heavy liquid) and magnetic separation techniques and then handpicked to an estimated visual purity of greater than 99.9 percent. Following sample preparation, the two mineral concentrates were loaded in aluminum foil capsules, sealed in silica vials, and irradiated for 30 h in the central thimble of the U.S. Geological Survey TRIGA reactor in Denver, Colorado. Vertical and horizontal gradients in neutron fluence in the irradiated package were monitored by placing 8–10 standards distributed along the length of each vial; the geometry of the irradiated package was such that each unknown sample was adjacent to at least one standard. Neutron flux was monitored using hornblende standard MMhb-1, which has a K-Ar age of 520.4 ± 1.7 Ma (Samson and Alexander, 1987). Corrections for reactor-produced interfering reactions were made using argon isotopes of K_2SO_4 and CaF_2 irradiated in each package.

After irradiation, the samples were progressively degassed in a double-vacuum resistance furnace in a series of eight to ten 20-min-long steps to a maximum temperature of 1450 °C. After each heating step, the gas was collected and purified using Zr-Al-Ti getters, and all five argon isotopes were measured using a mass spectrometer operated in the static mode. Apparent ages were calculated using decay constants recommended by Steiger and Jäger (1977). The argon data were evaluated using age spectra and $^{39}Ar/^{40}Ar$ versus $^{36}Ar/^{40}Ar$ correlation diagrams. The determination of whether the individual apparent ages in an age spectrum yielded a "plateau" was made using the criteria of Fleck et al. (1977). Following these criteria, a plateau is defined as comprising two or more contiguous gas fractions that yield apparent ages which are statistically indistinguishable at the 95% confidence level (using the critical value test of Dalrymple and Lanphere, 1969) and which together constitute greater than 50% of the total potassium-derived ^{39}Ar ($^{39}Ar_K$) released in the incremental heating experiment. Plateau ages were calculated using a weighted mean based on the proportion of $^{39}Ar_K$ released during the

incremental heating experiment. Total gas, plateau, and correlation ages are given at ±2σ and include the analytical uncertainty in the determination of the flux parameter *J*.

RESULTS

Paleomagnetism

Of the thirty sites collected for analysis, nine failed to yield usable paleomagnetic data. These samples were characterized by either erratic behavior during demagnetization, and thus failure to yield coherent linear or curvilinear behavior during the demagnetization experiments, or they yielded sample directions that were inconsistent at the site level. These samples are not considered further.

Samples from 21 sites yielded coherent paleomagnetic data during demagnetization experiments. For most cases, the demagnetization experiments yielded well-defined magnetizations of moderate to high coercivity (Fig. 3) and complete unblocking at temperatures below 580 °C. However, some samples show demagnetization trajectories in which stable endpoint behavior was not observed, yielding magnetizations that showed curvilinear demagnetization behavior during either AF or thermal demagnetization. In these samples, two or more magnetizations of overlapping coercivity and/or laboratory unblocking temperatures are present. Low coercivity and unblocking temperature magnetizations are similar to present-day field directions or appear to be randomly oriented. During progressive demagnetization these samples follow great circle trajectories toward the characteristic remanent magnetization defined by samples that yield stable endpoint behavior (Fig. 4). The thermal demagnetization data and temperature versus susceptibility curves are consistent with low-titanium magnetite as the principal remanence

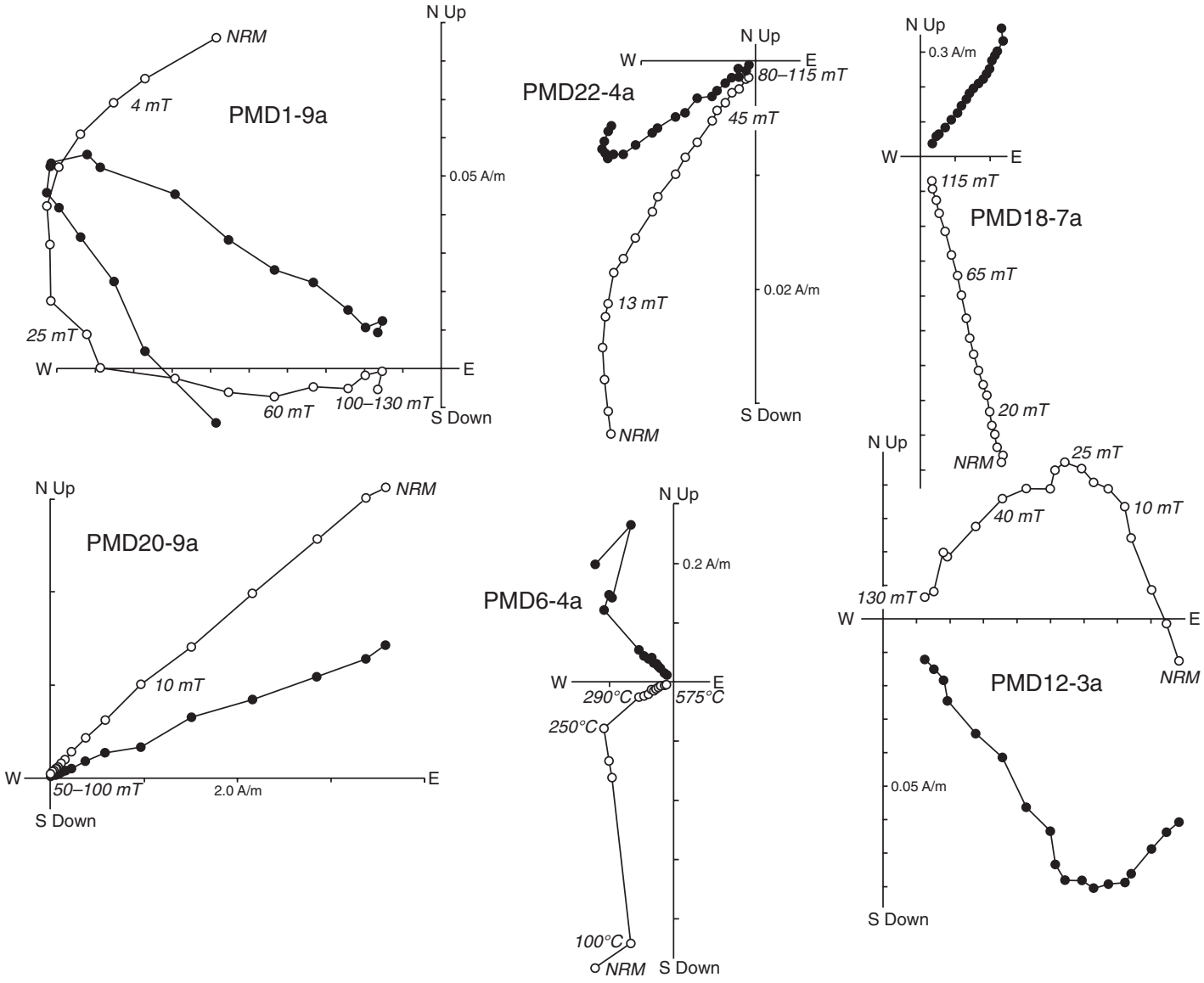


Figure 3. Representative in situ orthogonal vector diagrams of samples of intermediate composition sills of the Doherty Mountain fold complex that show well-defined isolation of characteristic remanent magnetizations during alternating field (AF) or thermal demagnetization. Solid circles are projections onto the horizontal plane, and open circles represent projections on the vertical plane. NRM—natural remanent magnetization at 22 °C.

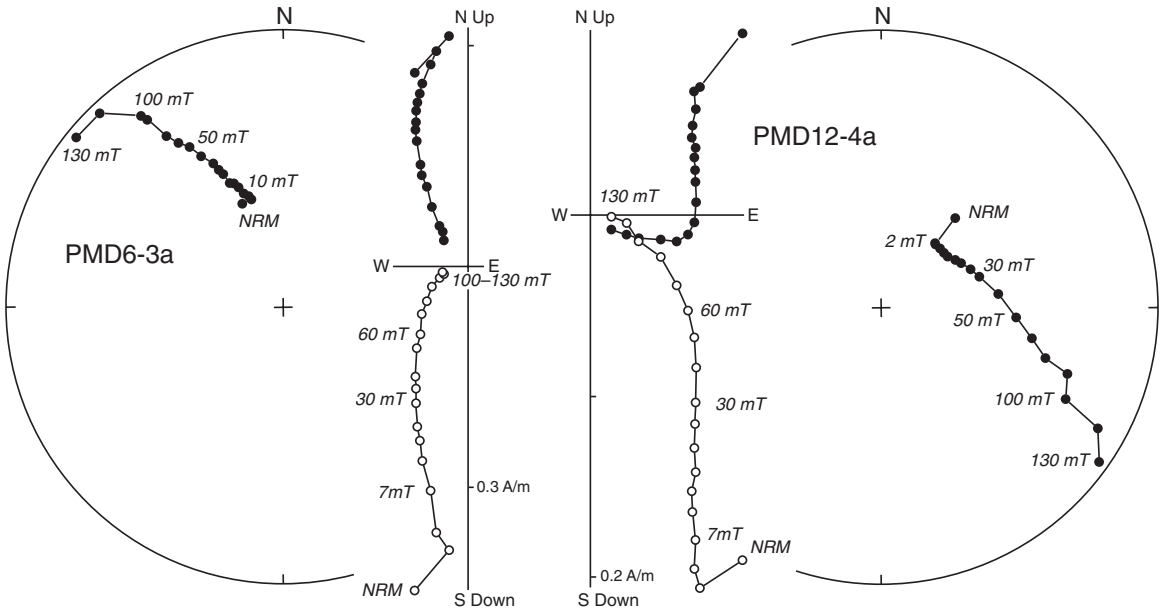


Figure 4. In situ orthogonal vector diagrams and equal area projections showing examples of degmagnetization behavior for samples in which the characteristic remanent magnetization could not be completely isolated because of its overlapping coercivity and/or laboratory unblocking temperature spectra with a well-defined secondary magnetization of northerly declination and moderate positive inclination. During progressive demagnetization, the magnetization vector follows a great circle trajectory toward the location of samples that yield well-defined stable endpoints during alternating field (AF) or thermal demagnetization. For the equal area projections, filled circles are projections on the lower hemisphere. Other conventions as in Figure 3.

carrier. Individual site-mean directions are well to moderately well grouped, with α_{95} values ranging from 3° to 14° and precision parameter (k) values of 477 to 16 (Table 1).

Rock magnetic experiments on selected samples from sills (Fig. 5) support our interpretation that magnetite is the principal remanence carrier in these rocks, although a higher coercivity phase (i.e., hematite) is present in considerable abundance at some sites and may carry part of the observed remanent magnetizations (e.g., PMD20-7a). IRM acquisition curves (Figs. 5A and 5B) vary in degree of downward concavity, suggesting a range of magnetite domain states is present. The modified Lowrie-Fuller test (Johnson et al., 1975), comparing AF demagnetization response of ARM and SIRM yields mixed results (Figs. 5C and 5D), with no clear indication of dominance by either multi-domain or single-domain magnetite states. Thermal demagnetization of three-component IRM (Lowrie, 1990) (Figs. 5E and 5F) shows that the assemblage of grains activated at 0.4 T dominates the signal, yet displays a mixed range of unblocking temperatures. In some cases, unblocking of the 0.4 T IRM occurs well below 580 °C, suggesting the presence of a high-titanium titanomagnetite as an important magnetic phase. The low-coercivity fraction (<0.1 T) is of limited intensity.

Progressive thermal demagnetization of samples from carbonate rocks at site CY shows remanence unblocking over a 280 to 330 °C temperature range, with complete unblocking occurring at temperatures between 350 and 360 °C (Figs. 6A and 6B). Alternating field demagnetization yields essentially univectorial decay of a well-defined moderate- to high-coercivity remanence in fields up to 120 mT. The combination of moderate to high coercivity and complete laboratory unblocking by temperatures less than 350 °C is consistent with the characteristic remanence of the samples being carried by pyrrhotite. The presence of pyrrhotite in marly limestones is typically the product of a secondary magnetization acquisition process that is induced during thermal metamorphism at temperatures greater than 200 °C (Crerar et al., 1978; Gillett et al., 1982; Gillett, 1991). In contrast, samples from site CX in the baked contact zone of one of the sills (PMD3) yield well-defined demagnetization data, with directions similar to those from the adjacent 1-m-wide sill, and significantly higher laboratory unblocking temperatures, showing complete unblocking below 580 °C (Fig. 6C). For these samples, the magnetic carrier is likely magnetite. At distances greater than 0.6 m, the host rocks yield unstable paleomagnetic behavior characterized by erratic directional and intensity changes at each demagnetization step, and thus do not carry a coherent remanent magnetization.

The carbonate strata exposed at site CY, well removed from any sill, and the baked contact rocks adjacent to site PMD 3 yield magnetizations that are identical to those obtained from the adjacent sills (Figs. 7A and 7B; Table 1). The directions obtained from the baked contact rocks at CY and the carbonate rocks at site CX yield in situ site-mean directions of west-northwest declination and shallow positive inclination. These directions are similar to expected directions for the middle Cambrian for North America (expected direction is Dec. = 276°, Inc. = 11°, α_{95} = 15°; calculated from data in Table 5.1 of Van der Voo [1993]); however, upon plunge and bedding tilt corrections, the observed directions plot close to the Late Cretaceous expected direction based on the Adel Mountain Volcanics pole of Gunderson and Sheriff (1991) (Figs. 7A and 7B). The baked contact test at site CX (sill site PMD3; Fig. 7A) suggests the remanence characteristic of the sill is likely a primary thermoremanent magnetization acquired during sill intrusion. However, the isolation of a remanence of similar direction in limestone host rocks of site CY, which is well removed from any intrusion, is problematic. Based on the similarity of

TABLE 1. PALEOMAGNETIC DATA FROM INTERMEDIATE AND MAFIC COMPOSITION SILLS IN THE DOHERTY MOUNTAIN FOLD COMPLEX, SOUTHWESTERN MONTANA

Site no.	Slat. (°N)	Slong. (°W)	In situ strike/dip	Plunge-corrected strike/dip	N/N_o	In situ		Structurally		*k*	α_{95}
						Dec.	Inc.	Dec.	Inc.		
Fold 1											
PMD19	45.9130°	111.9076°	290°/65°	345.3°/41.7°	11/11	207.6°	56.0°	345.8°	58.5°	137	3.7°
PMD20	45.9144°	111.9130°	276°/58°	357.1°/29.2°	8/9	196.0°	60.3°	339.2°	60.9°	52.6	7.2°
PMD21	45.9169°	111.9100°	206°/70°	335.2°/51.1°	8(2)/9	97.8°	56.5°	335.2°	51.1°	31.7	9.3°
PMD22	45.9156°	111.9207°	300°/70°	341.2°/51.7°	7/8	233.3°	48.1°	328.2°	56.2°	477	2.6°
PMD23[†]	45.9160°	111.9211°	298°/68°	343.1°/49.4°	8/9	341.0°	74.6°	345.1°	11.1°	76.9	5.9°
Fold 2											
PMD1	45.8781°	111.8931°	30°/90°	333.0°/137.5°	9(2)/11	307.2°	-10.6°	285.5°	77.2°	28.6	9.8°
PMD2	45.8774°	111.8923°	027°/87°	15.0°/129.0°	6(2)/7	309.2°	12.1°	17.6°	74.9°	62.2	8.6°
PMD3	45.8770°	111.8918°	003°/88°	354.8°/115.1°	8(4)/13	288.4°	10.3°	340.2°	74.3°	111	5.5°
PMD4	45.8837°	111.8902°	354°/88°	346.9°/10.7°	9(4)/10	272.4°	6.1°	323.1°	80.7°	106	5.9°
PMD6	45.8884°	111.8883°	030°/110°	354.6°/148.2°	9(6)/10	312.5	-16.1°	319.6°	77.5°	50.1	7.3°
PMD10	45.8953°	111.8887°	210°/85°	191.7°/42.7°	9(7)/9	88.0°	12.3°	34.1°	57.6°	69.9	6.2°
PMD12	45.8924°	111.8928°	232°/97°	228.3°/44.9°	12(7)/13	131.5°	-5.3°	49.8°	79.4°	40.2	6.4°
PMD15	45.8895°	111.8969°	252°/75°	267.8°/24.0°	10/11	135.1°	41.0°	12.5°	55.1°	38.1	7.5°
CX	45.8780°	111.8933°	005°/90°	353.1°/116.5°	7/15	291.5	14.0°	346.6°	68.5°	179	4.2°
CY	45.8767°	111.8912°	010°/91°	356.1°/120.9°	12/12	293.5°	11.1°	352.2°	71.9°	114	3.9°
Fold 3											
PMD24	45.9309°	111.8611°	218°/36°	155.0°/39.3°	9/10	66.2°	67.3°	1.2°	58.6°	25.1	9.9°
PMD25	45.9322°	111.8540°	340°/90°	347.4°/69.7°	7/10	271.0°	18.9°	351.2°	62.0°	42.1	8.7°
PMD26	45.9307°	111.8528°	346°/95°	348.9°/77.4°	5/10	294.3°	11.3°	331.6°	48.6°	30.0	14.2°
Fold 4	Includes site PMD24										
PMD27	45.9319°	111.8792°	320°/49°	353.0°/37.2°	10/10	252.4°	51.9°	352.5°	71.2°	148	3.8°
PMD28	45.9230°	111.8806°	320°/49°	353.0°/37.2°	10/10	260.4°	56.9°	358.9°	64.8°	54.2	6.3°
PMD30	45.9289°	111.8775°	326°/46°	0.9°/37.6	9(1)/10	260.6°	54.7°	358.0°	70.9°	16.0	13.7°

Note: Site designation on Fig. 2. Site no.—sample site designation. Slat. and Slong.—latitude and longitude of the site, respectively. Strike/dip—in situ strike and dip of host strata (strike is given as 90° clockwise of the dip direction). Plunge-corrected strike/dip is the modified strike and dip after correction for fold plunge. N/N_o—number of samples used in the site-mean calculation versus those analyzed (value in parentheses is the number of great circles used to calculate the mean where both linear and great circle data have been used). Dec. and Inc.—site-mean declination and inclination before and after application of plunge and tilt corrections. *k*—Fisher's (1953) precision parameter; α_{95}—the semi-angle of the 95% confidence level about the site-mean direction. CX—Baked contact test for paleomagnetic site PMD3; CY—carbonate host rock samples.

[†]The characteristic magnetization of site PMD23 appears to postdate folding. Consequently, it is treated as an independent reading of the geomagnetic field, as discussed in the text.

the ChRM directions (Fig. 7B) and the presence of pyrrhotite in these rocks, we suggest that the ChRM of site CY is a secondary magnetization probably caused by thermal effects associated with intrusion of the numerous sills in close proximity to the sample location.

For all four folds, in situ site-mean directions are scattered, but most sites become increasingly well grouped following application of plunge and bedding tilt corrections as judged by the increase of the precision parameter (*k*) with unfolding (Fig. 8). An exception is site PMD23 from Fold 1 whose in situ direction (Dec. = 341°, Inc. = 75°, α_{95} = 5.9°; Table 1) is essentially identical to Late Cretaceous and/or early Tertiary reference directions. Application of plunge and tilt corrections to this site causes the site-mean direction to move in a direction opposite to the other three sites from this fold (Fig. 8). The reason for this observation is unknown, but could be due to complete remagnetization of this site subsequent to folding. Alternatively, because this site is compositionally distinct from all other sites sampled in this fold, site PMD23 could represent part of a different sill emplaced at a younger time (i.e., subsequent to folding). Because the magnetization recorded by this site appears to postdate folding, it has been excluded from the fold test results discussed below.

The majority of site-mean directions from the Doherty Mountain sills clearly show better grouping following application of plunge and tilt corrections, thus suggesting that their characteristic remanent magnetizations predate folding. For two of the folds (Fold 1 and Fold 4), the optimal degree of unfolding determined by the Watson and Enkin (1993) optimal concentration method is essentially 100 percent (Table 2; Fig. 8). However, for two folds (Fold 2 and Fold 3), the optimal concentration occurs at slightly higher values (i.e., ~112 percent). The Watson and Enkin (1993) method is thus sensitive to the assumption that all sources of uncertainty have been adequately sampled, which is unlikely in studies such as this where the data are derived from the relatively small number of sills sampled in these folds. Values in excess of 100 percent could result from several factors, including uncertainties in appropriate tilt axes used to restore the data, slightly different ages of magnetization (especially if sills of slightly different age were sampled), inadequate sampling and averaging of paleomagnetic secular variation, and failure to fully isolate the characteristic remanent magnetization of each site (Watson and Enkin, 1993). Thus, the confidence interval for the degree of optimal concentration above the 100% level is underestimated. Accordingly, we use the tilt corrected values at 100% unfolding as the best estimate of the concentration of

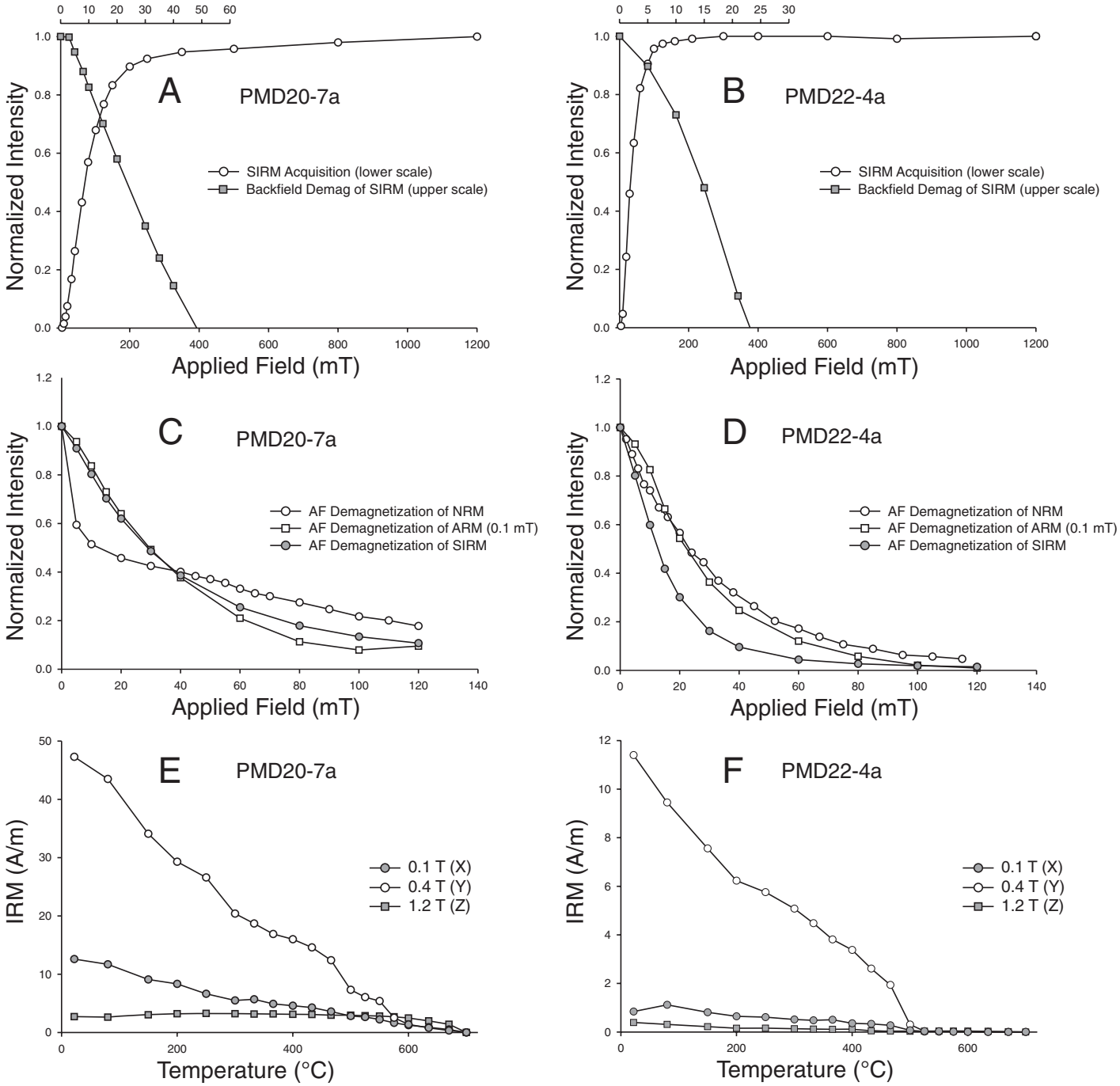


Figure 5. Results of rock magnetic experiments from intrusive rock samples from the Doherty Mountain fold complex. (A, B) acquisition of a saturation isothermal remanent (SIRM) magnetization (lower scale) and direct current demagnetization of a backfield SIRM (lower scale); (C, D) comparison of alternating field (AF) demagnetization curves for the natural remanent magnetization (NRM), anhysteretic remanent magnetization (ARM), and SIRM; (E, F) thermal demagnetization of three-axis isothermal remanent magnetization (IRM) (Lowrie, 1990) acquired in three orthogonal directions in fields of 0.1, 0.4, and 1.2 Tesla.

the site means. The optimal concentration of k at values on the order of 100 percent indicates that the remanence in most sills predates folding. In addition to the optimal concentration method, we also applied the direction-correlation test of Enkin (2003) to our fold test analyses. For all four folds, the D.C. test yields values that are indistinguishable from 100 percent untilting. On the basis of the fold test data, we suggest that the tests are positive and indicate that the ChRM of the sills predates folding. Assuming that the ChRM observed in the sills is a primary magnetization acquired during intrusion, our results suggest that sill intrusion must have predated folding. Alternatively, given that the carbonate rocks at site CY yield a corrected site-mean direction similar to those of the sills in this study, it is possible that the remanence observed in both the

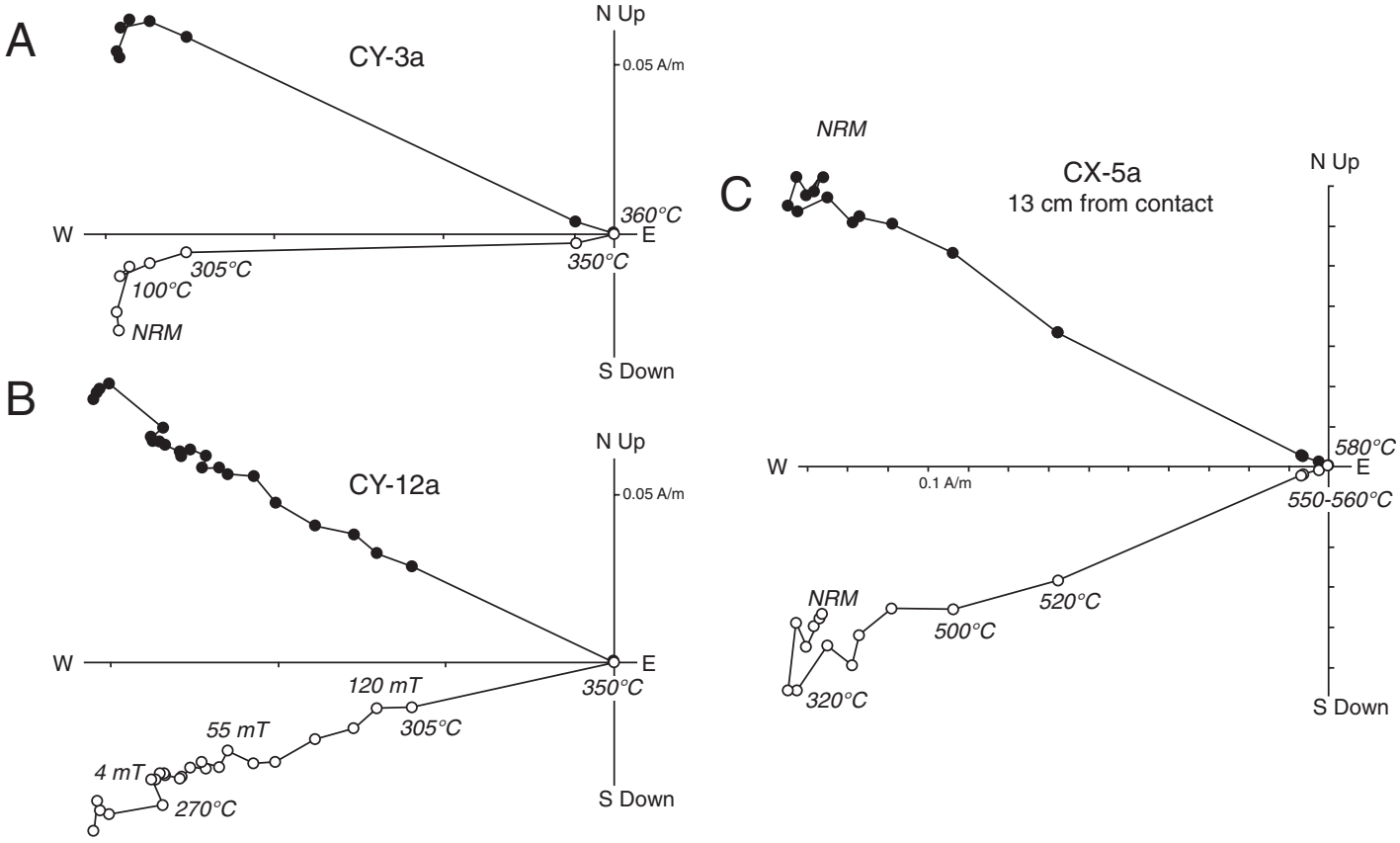


Figure 6. In situ orthogonal vector diagrams showing demagnetization behavior of samples from Paleozoic carbonate rocks. (A) carbonate rocks from within 13 cm of an ~1-m-thick sill. (B) and (C) vector diagrams from apparently remagnetized carbonate rocks originally thought to be well removed from thermal effects associated with sill intrusion as described in the text. Conventions as in Figure 3.

sills and carbonate rocks is secondary and due to complete remagnetization of these rocks, prior to folding, associated with widespread latest Cretaceous igneous activity in southwest Montana, which may have spanned the time period prior to, during, and after folding. Although we favor the interpretation that the remanence in most of the sills is a primary thermoremanent magnetization acquired during sill emplacement and cooling, it is not possible to distinguish between these alternatives given existing data. Regardless, the plunge-and-tilt–corrected mean directions obtained from the Doherty Mountain sills are similar to the expected direction (Dec. = 352°, Inc. = 63°) for the ca. 75 Ma reference pole obtained from the 76–73 Ma Adel Mountain Volcanics of western Montana (Gunderson and Sheriff, 1991; Harlan et al., 2005).

$^{40}Ar/^{39}Ar$ Geochronology

Biotite from two different fold localities yield straightforward $^{40}Ar/^{39}Ar$ results, yielding total gas ages of 76.85 ± 0.32 Ma and 76.27 ± 0.32 Ma (both errors at 2σ), respectively (GSA Data Repository Table DR1[1]). The age spectra from both samples (Fig. 9) show slight discordance, but are characterized by consistent ages over much of the $^{39}Ar_K$ released in the step-heating experiments. Calculation of weighted-mean plateau ages gives apparent ages of 77.18 ± 0.31 and 77.00 ± 0.31 Ma. Calculations of inverse isochrons give essentially ages, with atmospheric intercepts that are statistically identical to atmospheric $^{40}Ar/^{36}Ar$ values. Assuming these dates record cooling following sill emplacement, the dates indicate that sill intrusion occurred at 77 Ma; the normal polarity of the sills is consistent with magnetization acquisition during polarity chron C33n. The 77 Ma dates reported here are similar to isotopic dates from a lamprophyre dike from the Golden Sunlight mine 9 km west of the our field area (DeWitt et al., 1996. Biotite from the dike yielded a conventional K-Ar date of 79.8 ± 2.8 (2σ) and a 76.9 ± 0.5 Ma (2σ) $^{40}Ar/^{39}Ar$ plateau date. The 77 Ma dates are also similar to a recently obtained U-Pb date of 74.5 Ma from an intrusion in the Doherty Mountain area (J.B. Mahoney, 2007, written commun.). On a regional scale, the dates are consistent with recently reported sensitive high-resolution ion microprobe (SHRIMP) U-Pb and $^{40}Ar/^{39}Ar$ dates from the nearby Boulder Batholith and satellite plutons that range from 80.4 to 74.5 Ma (Lund et al., 2002).

[1]GSA Data Repository Item 2008120, $^{40}Ar/^{39}Ar$ analytical data for Doherty Mountain sills, is available at www.geosociety.org/pubs/ft2008.htm. Requests may also be sent to editing@geosociety.org.

DISCUSSION

The ubiquitous presence of fossiliferous, readily dated synorogenic sedimentary rocks in much of the Cordilleran thrust belt farther south in Idaho, Utah, and Wyoming has allowed the timing of the individual thrust movements in these areas to be well refined (Dorr et al., 1977; Wiltschko and Dorr, 1983; DeCelles et al., 1995; Constenius, 1996). In northwest Wyoming, the timing of local, thin-skinned and basement-involved deformation has also been well determined based on key synorogenic strata whose ages have been paleontologically defined (Dorr

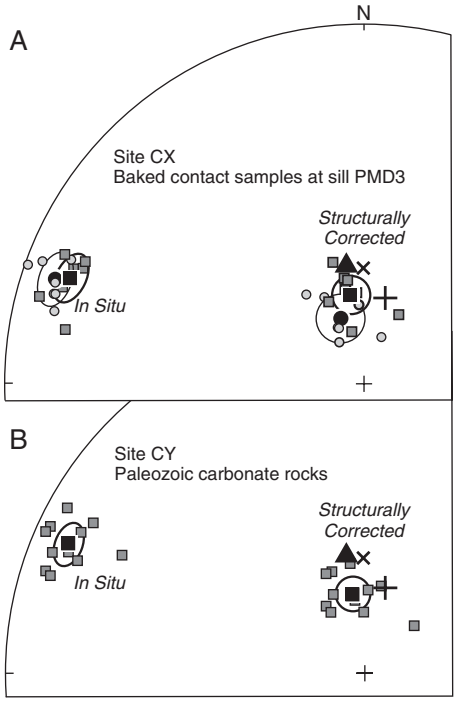


Figure 7. (A) Equal-area projection showing results of a paleomagnetic baked contact test showing individual sample directions (squares) obtained from carbonate rocks within ~10–60 cm of an ~1-m-thick sill and their site-mean direction (large square) and its α_{95} cone of confidence. Also shown are sample directions (circles) and site-mean and α_{95} obtained from the sill. The site-mean directions from the sill and the baked contact zone rocks are indistinguishable at the 95% confidence level. Beyond ~60 cm, carbonate rocks display inconsistent and unstable paleomagnetic behavior during demagnetization. (B) Sample- and site-mean directions from carbonate rocks thought to be well removed from thermal effects of sill intrusion. However, similarity of sample- and site-mean directions with those of nearby sills suggests that these Paleozoic rocks were remagnetized coincident with sill emplacement. Sample- and site-mean directions are shown in situ and after correction for plunge and bedding tilt as described in the text. Also shown are the 75 Ma reference direction from the Adel Mountain Volcanics pole (triangle), time-averaged direction for the study area (×), and present-day field direction (+).

et al., 1977). Movements on individual thrusts have been pinned to within a million years in some cases and rates of thrust movement to within a few mm/yr. In southwest Montana, timing of foreland deformation is reasonably estimated because of the presence of synorogenic conglomerates (e.g., Perry et al., 1988; Tysdal, 1986; DeCelles et al., 1987). However, timing of thrust fault movement in the Montana part of the Cordilleran thrust belt is an order of magnitude less well known than farther to the south. Although it might be argued that development of the Montana thrust belt occurred at the same time and at the same rate as contraction elsewhere, it is just as likely or more probable that thrust displacement and attending crustal shortening in the fold-and-thrust belt was diachronous.

The paleomagnetic data from 20 of 21 sites in sills from fourfolds studied in the Doherty Mountain fold complex show that each incremental fold test is consistent with maximum clustering at 100 percent unfolding. This indicates that remanence acquisition in most sites occurred prior to folding of the host sedimentary rocks. Accordingly, we interpret the data to support the hypothesis that sill intrusion at 77 Ma predated fold and thrust deformation in this area. Alternatively, we note that the remanence could reflect remagnetization of the sills and host rock, prior to folding, due to regional tectonothermal effects associated with Late Cretaceous magmatism and sill emplacement. Regardless, these data indicate that remanence acquisition for most of the sites in this study occurred at ca. 77 Ma, prior to folding of the host strata during fold-and-thrust deformation. Similarly, sills exposed in the folds of the Devils Fence anticline (Klepper et al., 1957; Mahoney et al., 2008), exposed north of the study area, probably also predate folding. However, given that isotopic dates in the area indicate that magmatic activity extends to as young 74 Ma, it is possible that some intrusions elsewhere in the Doherty Mountain and surrounding areas may be synchronous with, or postdate folding, although such relationships have not been documented. The identification of syn- and post-folding intrusions, combined with high-precision dating, could provide an important lower limit for the age of compressional deformation in this part of the fold and thrust belt.

Previous work on the timing of folding and thrusting in the Helena salient of the Montana thrust belt and adjacent foreland (Harlan et al., 1988; Schmidt et al., 1990; Burton et al., 1995; Lageson et al., 2001) concluded that shortening probably occurred between ca. 74 and 66 Ma. Intrusion of the Boulder Batholith and extrusion of associated volcanic rocks occurred over a more restricted time interval than shortening in the Helena salient, as constrained by U-Pb SHRIMP, $^{40}Ar/^{39}Ar$, and K-Ar age determinations reported by Robinson et al. (1968), Tilling et al. (1968), Vitaliano et al. (1979), and Lund et al. (2002) on phases of the Boulder Batholith and deformed volcanic rocks of the Elkhorn Mountains Volcanics. On a regional scale, thrusting probably began west of the batholith between 105 and 85 Ma (e.g., Hyndman et al., 1988) and ceased 100 km east of the batholith prior to 51 Ma (Harlan et al., 1988; Harlan, 2006; du Bray et al., 2006). Our data from the Doherty Mountain fold complex demonstrate that folding and thrusting in this area occurred subsequent to 77 Ma, based on the observation that most of the sills in the Doherty Mountain fold complex have been folded. This is consistent with results of Diehl (1991) who showed that rocks of the 78–74 Ma Elkhorn Mountains volcanics of the Bull Mountain area, 10 km northwest of our study area, were also significantly tilted during fold and thrust deformation of the Helena salient. Farther east as well as south of the Helena salient, the development of foreland folds and faults is considered to be coeval with the development of thrust belt structures, but crosscutting relationships indicate that the earliest movements on NW-trending basement-involved thrust faults occurred before movement on thin-skinned thrusts (Schmidt et al., 1988) and therefore before 77 Ma. Intrusion of the Tobacco Root batholith was coeval with movement on three of these NW-trending basement faults (Bismark, Mammoth, and Carmichael faults) (Schmidt et al., 1990). In the northern Madison Range, 60 km south of our study, Kellogg and Harlan (2007) have shown that emplacement of ca. 69 Ma porphyritic dacite sills postdates Laramide deformation in that area. In general, basement-involved faulting in the foreland overlaps in timing with thin-skinned thrusting, and, like the thin-skinned thrust faults, basement-involved deformation youngs toward the east. The youngest basement-involved faulting resulting in folding of thin-skinned thrusts in the Bridger Range is latest Paleocene or early Eocene time (Harlan et al., 1988; Lageson, 1989).

On a regional scale, two paleomagnetic studies have documented the presence and importance of statistically significant vertical-axis rotations of thrust sheets within and adjacent to the Helena salient during Laramide-age deformation. Eldredge and Van der Voo (1988) studied red beds of the Lower Cretaceous Kootenai Formation at several localities within the salient and in the southwest Montana reentrant. Their results indicated a complex and highly variable pattern of rotation for rocks of the thrust belt, which they ascribed to the effects of buttressing of thrust sheets due to interaction with crystalline basement of the Montana Foreland. Jolly and Sheriff (1992) presented paleomagnetic data from welded ash flow tuffs, latite flows, and interbedded sedimentary rocks from the Upper Cretaceous Two Medicine Formation northwest of the Helena salient near Wolf Creek, Montana. Based on the observation of a statistically significant (24.2° ± 14.0°) counterclockwise rotation when compared to

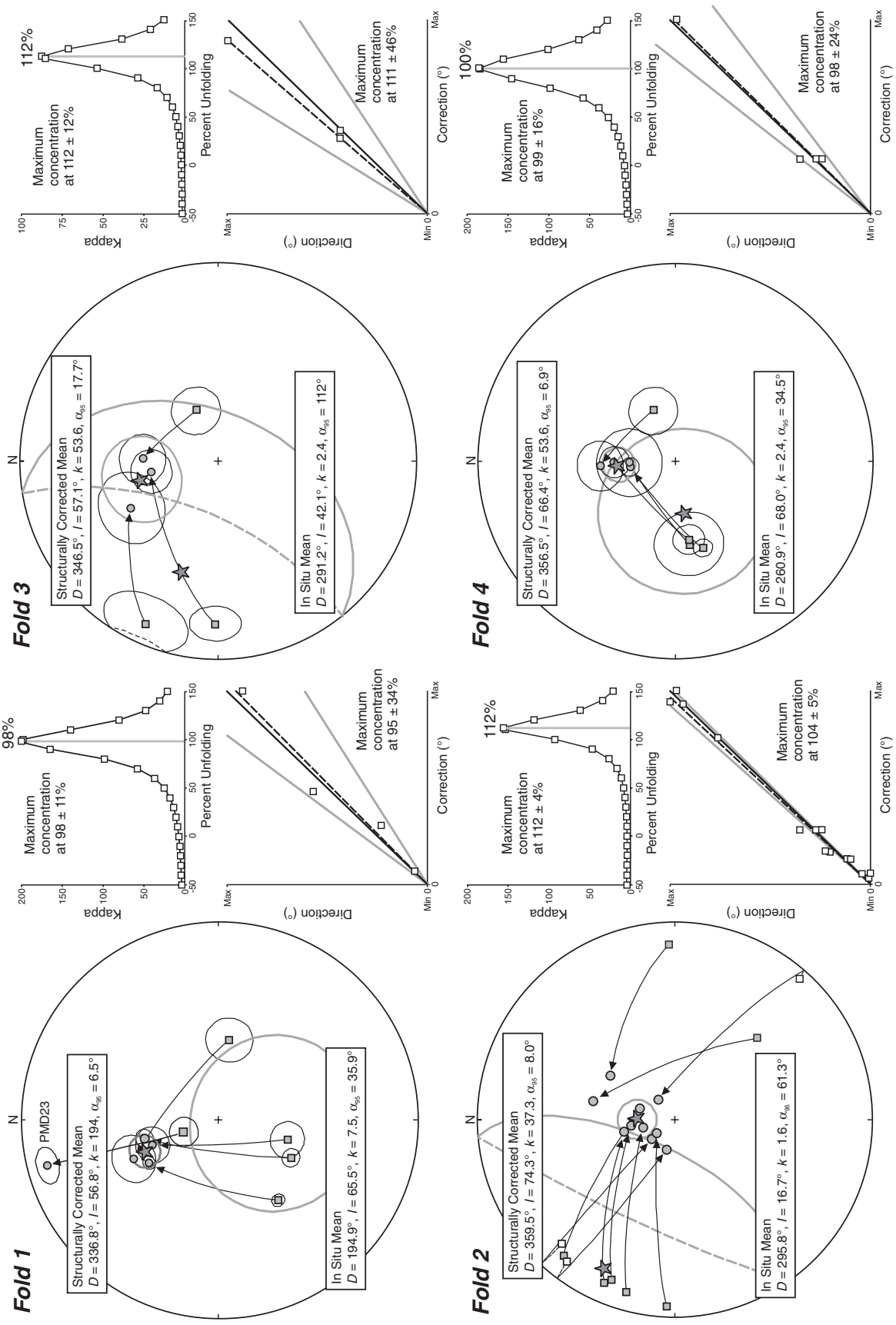


Figure 8. Equal-area projections showing the results of fold tests on individual folds, as described in the text. Circles show individual site-mean directions and their α_{95} cones of confidence. Stars indicate group-mean directions with their α_{95} cones of confidence (for ease of viewing, the α_{95} cones of confidence have been omitted from individual site-mean directions for Fold 2). Solid symbols are projections on the lower hemisphere; open circles are projections on the upper hemisphere. Solid lines with arrows show the trajectory of site-mean directions during progressive unfolding from the in situ to fully plunge- and tilt-corrected configurations. Also shown are results of incremental unfolding for each fold test as indicated by changes in Fisher's precision parameter (k) versus percent unfolding based on numerical simulation using the Watson and Enkin (1993) optimal concentration method and the maximum clustering of paleomagnetic sites using the Enkin (2003) direction-correlation test. Fold tests were performed by first removing effects of fold plunge and then applying tilt corrections. Note that for the fold test analysis applied to Fold 1, paleomagnetic results from site PMD23 have not been used as discussed in the text.

TABLE 2. FOLD TEST RESULTS FROM THE DOHERTY MOUNTAIN FOLD COMPLEX, SOUTHWESTERN MONTANA

		Dec.	Inc.	*k*	α_{95}	N	Optimal unfolding O.C.	Optimal unfolding D.C.
Fold 1	In situ	194.9°	65.5°	7.5	35.9°	4		
	Plunge correction	312.9°	41.1°	7.5	35.9°			
	Plunge and tilt corrected	336.8°	56.8°	194	6.6°		98 ± 11%	95 ± 34%
Fold 2	In situ	295.8°	16.7°	1.6	61.3°	10		
	Plunge correction	292.8°	-26.8°	1.6	61.3°			
	Plunge and tilt corrected	359.5°	74.3°	37.3	8.0°		112 ± 4%	104 ± 5%
Fold 3	In situ	291.2°	42.1	2.4	111.9°	3		
	Plunge correction	318.1°	22.5°	2.4	111.9°			
	Plunge and tilt corrected	346.5°	57.1°	53.6	17.7°		112 ± 12%	111 ± 46%
Fold 4	In situ	260.9°	68.0°	8.1	34.5°	4		
	Plunge correction	317.6°	54.9°	8.1	34.5°			
	Plunge and tilt corrected	356.5°	66.4°	179	6.9°		99 ± 16%	98 ± 24%

Note: Dec. and Inc.—site-mean declination and inclination; *k*—Fisher's (1953) precision parameter; α_{95}—the semi-angle of the 95% confidence level about the site-mean direction; N—number of sites used in the paleomagnetic fold test. Optimal unfolding O.C. is the percentage of unfolding that maximizes *k* during numerical simulation based on the method of Watson and Enkin (1993). Optimal unfolding D.C. is the best estimate of site clustering with unfolding based on the direction-correlation test of Enkin (2003).

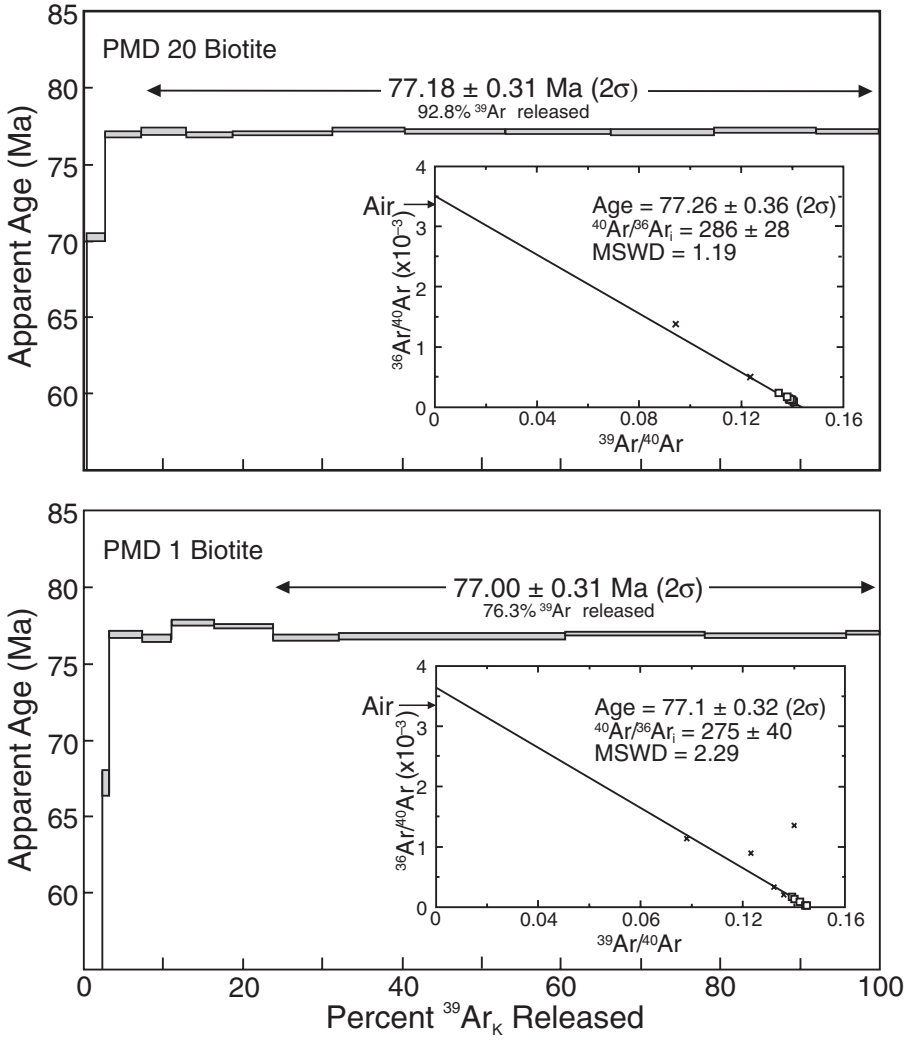


Figure 9. $^{40}Ar/^{39}Ar$ age spectra for biotite concentrates from two different sills in the Doherty Mountain fold complex. The height of the rectangle for each temperature step represents the analytical error in the apparent age at ±1σ. The error in the plateau dates is reported at ±2σ. Insets in each diagram show the results of $^{36}Ar/^{40}Ar$ versus $^{39}Ar/^{40}Ar$ inverse isochron analysis for each sample. Squares show steps used in defining the isochron; crosses are steps that were excluded from the isochron analysis. MSWD—mean square of weighted deviates.

the Late Cretaceous Adel Mountain Volcanics reference pole, they suggested that rotation of the Two Medicine rocks was due to left-lateral movement within thrust sheets as they interacted with crystalline basement of the foreland. Thus, the paleomagnetic evidence from these two studies suggests that significant vertical-axis rotations may be a part of the kinematic evolution of thrust sheets within and near the Helena salient. Indeed, vertical-axis rotations associated with the three-dimensional kinematic evolution of curved orogenic belts have been documented by many researchers (c.f. Sussman et al., 2004; Weil and Sussman, 2004; Kwon and Mitra, 2004).

Although our data set is limited in terms of independent recordings of the ancient geomagnetic field and probably does not fully average secular variation, we compare the paleomagnetic data from the Doherty Mountain fold complex with expected time-averaged results for North America. Our calculation is based on the mean directions obtained from the four fold tests after correction for fold plunge and 100% untilting. In addition, we include a fifth paleomagnetic site, PMD23 (Table 1), whose in situ site-mean direction is similar to that expected for rocks of Late Cretaceous age and appears to have not been tilted significantly. Averaging these five observations, we obtain a grand-mean direction of Dec. = 347°, Inc. = 66° (k = 71.9, α_{95} = 9°). The overall mean result probably accounts for at least eight independent observations of the geomagnetic field, because multiple sills were sampled in Fold 2, and the carbonate rocks sampled at CY may represent a slightly different record of the field due to their lower laboratory unblocking temperatures. Our mean direction is statistically indistinguishable from the expected directions for the study area derived from the reference pole from the 75 Ma Adel Mountain Volcanics (Gunderson and Sherriff, 1991) or the ca. 80 Ma reference North American reference pole of Harrison and Lindh (1982) (Fig. 10). We interpret this result to suggest that the Doherty Mountain area and this part of the Southwest Montana Transverse Zone have not experienced significant vertical-axis rotations associated with the interactions of the thrust sheets of the Helena salient with the adjacent Rocky Mountain foreland uplifts. This result is consistent with the results reported by Eldredge and Van der Voo (1988), whose data also did not support evidence for rotations in this region, and to structurally corrected paleomagnetic results from folded volcanic rocks of the Bull Mountain section of the Elkhorn Mountains volcanics (Dec. = 342°, Inc. = 67°, k = 47.2, α_{95} = 7.5°, N = 10 sites; reverse polarity sites inverted to normal polarity) reported by Diehl (1991) along the southern margin of the salient east of our study area. Although

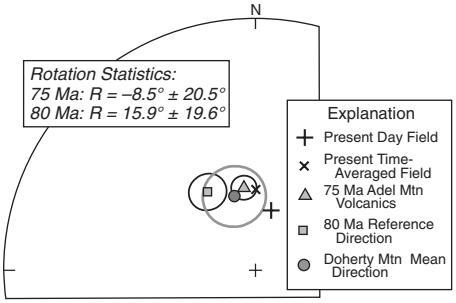


Figure 10. Equal-area lower hemisphere projection showing the grand-mean direction from the sills of the Doherty Mountain fold complex sampled in this study versus Late Cretaceous reference (Gunderson and Sheriff, 1991; Harrison and Lindh, 1982) directions for the study area. Also shown are the α_{95} cones of confidence for the directions shown in the projection and associated rotation statistics based on comparison of the Doherty Mountain mean direction and the reference directions.

the paleomagnetic data from the intrusive rocks of the Doherty Mountain fold complex are consistent with expected cratonic reference directions and studies from elsewhere in the Helena salient, these results must be viewed with caution. The number of independent observations of the geomagnetic field is low, and the sites are only of normal polarity. Thus, there is the distinct possibility that the results do not fully average paleosecular variation. Also, as noted previously, failure to fully correct for fold plunge in restoring rocks to the paleohorizontal can result in significant apparent rotations that could influence our results. However, the consistency of results from three different studies suggests that appreciable vertical-axis rotations due to thrust sheet movement do not seem to have played a role in the structural evolution of this part of the Helena salient of the Cordilleran fold and thrust belt. Finally, although it is tempting to use the paleomagnetic data from the Helena salient to refine the three-dimensional evolution of this part of the Cordilleran thrust belt in terms of displacement field models for curved orogenic belts (i.e., Sussman et al., 2004), existing data within the salient are too sparse, do not have sufficient areal distribution, and are not robust enough to make meaningful interpretations regarding the strain history of the salient.

CONCLUSIONS

Paleomagnetic incremental fold test results from diorite sills in the Doherty Mountain area of southwest Montana indicate that they were emplaced prior to fold-and-thrust belt deformation. Biotite from two samples from different sills at two localities gives identical apparent $^{40}Ar/^{39}Ar$ age spectra, with dates of ca. 77 Ma. These values are broadly correlative with nearby intrusions of the Late Cretaceous Boulder Batholith and its satellite plutons and the Elkhorn Mountains Volcanics. The paleomagnetic and geochronologic data indicate that the thin-skinned fold-and-thrust structures in the Doherty Mountain fold complex postdate the 77 Ma (Campanian-Maestrichtian) magnetization recorded in the sills. The paleomagnetic data presented here are consistent with other paleomagnetic studies along the southern margin of the Helena salient that have shown that this area has not been affected by significant local vertical-axis rotations associated with impingement of thin-skinned folds and faults of the thrust belt with coeval foreland structures to the south unlike results from elsewhere along the margins of the Helena salient and in the McCarthy Mountain reentrant to the south (Eldredge and Van der Voo, 1988; Jolly and Sheriff, 1992).

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